

MADI Project: Theoretical study and Practical implementation of FCI (Fast Causal Inference) Algorithm

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1 Introduction

In the era of Artificial Intelligence and Modern Statistics, the study of causality between features within a database plays a pivotal role in knowledge acquisition. While the rapid development of Machine Learning and Deep Learning allows for effective problem modeling through probabilistic and statistical formulas, the challenge of logical inference remains. Purely empirical correlations obtained by standard models often fail to distinguish between mere association and rational causal conclusions.

The prices of empiricism are primarily driven by two factors: "latent variables" — unobserved factors that influence the visible variables but are absent from the dataset — and "selection bias" — distortions arising from the data collection process itself [1]. Consequently, Causal Inference has emerged as a crucial field. Initially popularized through Judea Pearl's *Probabilistic Reasoning in Intelligent Systems* (1988) [2], it was further advanced by Spirtes, Glymour, and Scheines with the PC (Peter-Clark) and FCI (Fast Causal Inference) algorithms, as detailed in their seminal work *Causation, Prediction, and Search* (1990).

As a Master's project, this work aims to understand the principles of the FCI algorithm and implement it using the `pyagrum` library. The project involves a theoretical review, the practical implementation of the algorithm to handle latent confounders, and a comparative analysis against other causal discovery methods.

2 Theoretical study

2.1 Definitions and Notions

1. Partial Ancestral Graph:

A Partial Ancestral Graph (PAG) is defined as a triplet

$$\mathcal{G} = \langle \mathcal{V}, \mathcal{E}, \mathcal{M} \rangle,$$

where:

- $\mathcal{V} = \{X_i\}_{i=1}^n$ is the set of n nodes, each representing a distinct variable.
- $\mathcal{E} \subseteq \{\{X_i, X_j\} \mid \{X_i, X_j\} \subseteq \mathcal{V}\}$ is the set of (unordered) edges.
- $\mathcal{M} = (m_{ij})_{i=1, j=1}^{i=n, j=n}$ is the edge-mark matrix, where

$$m_{ij} \in \{\varepsilon, -, \circ, >\}.$$

The value $m_{ij} = \varepsilon$ indicates the absence of an edge mark at the endpoint of X_j on the edge incident to X_i . Whenever $m_{ij} \neq \varepsilon$, we may write $m_{ij} = *$ to denote the presence of an edge mark.

At initialization (step 0), the graph is defined as

$$\mathcal{G}^0 = \langle \mathcal{V}, \mathcal{E}^0, \mathcal{M}^0 \rangle,$$

where:

- $\mathcal{E}^0 = \{\{X_i, X_j\} \mid \{X_i, X_j\} \subseteq \mathcal{V}\}$, i.e., the initial graph is complete.

- $\mathcal{M}^0 = (m_{ij}^0)_{i=1, j=1}^{i=n, j=n}$, with

$$m_{ij}^0 = \begin{cases} \varepsilon, & \text{if } \{X_i, X_j\} \notin \mathcal{E}^0, \\ \circ, & \text{if } \{X_i, X_j\} \in \mathcal{E}^0. \end{cases}$$

The "grammar" of every edge is convened as follow:

$$\forall i \forall j (\{X_i, X_j\} \subseteq \mathcal{V} \implies (\{X_i, X_j\} \in \mathcal{E} \iff X_i[m_{ji}] - [m_{ij}]X_j))$$

2. Relations between nodes

- Parenthood:** A node A is called **parent** of a node B (B is called **child** of A) if and only if $A \rightarrow B$.
- Unshielded triplet:** For all triplet being subset of $\mathcal{V}$, a triplet $\langle A, B, C \rangle$ is **unshielded** if $\{A, B\} \in \mathcal{E}$, $\{B, C\} \in \mathcal{E}$ and $\{A, C\} \notin \mathcal{E}$.
- Collider:** For all unshielded triplet $\langle A, B, C \rangle$ in $\mathcal{G}$, B is considered a **collider** if $A * \rightarrow B \leftarrow * C$.
- Triangle:** A triplet $\langle A, B, C \rangle$ forms a triangle if $\{A, B\} \in \mathcal{E}$, $\{B, C\} \in \mathcal{E}$ and $\{A, C\} \in \mathcal{E}$.
- Circle edge:** For $\{A, B\} \in \mathcal{E}$, $\{A, B\}$ is called a **circle edge** if $A \circ - \circ B$.
- Potentially directed edge:** For $\{X_i, X_j\} \in \mathcal{E}$, $\{X_i, X_j\}$ is called potentially directed if $m_{ji} \neq >$ and $m_{ij} \neq -$.

3. Paths

We consider a path $\pi = (\pi_i)_i \in \mathcal{E}^*$ among $\mathcal{G}$. The direction of edges is ignored.

- Inducing path:**

$\pi = (\pi_k)_{k=1}^l$ is an **inducing path** if for all triplet $\langle \pi_{k-1}, \pi_k, \pi_{k+1} \rangle$ in π , either $\pi_{k-1} * \rightarrow \pi_k \leftarrow * \pi_{k+1}$ or $\langle \pi_{k-1}, \pi_k, \pi_{k+1} \rangle$ forms a triangle.

- Discriminating path:**

$\pi = (\pi_k)_{k=1}^l$ is a **discriminating path** for π_{l-1} if:

- π contains at least three edges.
- There is no edge between π_1 and π_l .
- $\forall \pi_k \left(\pi_k \in \{\pi_k\}_{k=2}^{l-2} \implies (\pi_{k-1} * \rightarrow \pi_k \leftarrow * \pi_{k+1} \wedge \pi_k \rightarrow \pi_l) \right)$

- Uncovered path:**

$\pi = (\pi_k)_{k=1}^l$ is an **uncovered path** if all triplet $\langle \pi_{k-1}, \pi_k, \pi_{k+1} \rangle$ triplets in π are unshielded.

- An uncovered path where all edges are circle is an **uncovered circle path**.
- An uncovered path where all edges are potentially directed is an **uncovered potentially directed path**.

4. **Perfect Causal Graph:** A perfect Bayesian network, in which the causalities between nodes are clearly defined, must be a Directed Acyclic Graph (DAG). Considering $\mathcal{G}_{\text{DAG}} = \langle \mathcal{V}, \mathcal{E}_{\text{DAG}}, \mathcal{M}_{\text{DAG}} \rangle$ as the obtained output, its properties are as follow:

- Every edges are strictly oriented:

$$\forall i \forall j (\{X_i, X_j\} \in \mathcal{E}_{\text{DAG}} \implies (m_{ij} = > \iff m_{ji} = -))$$

- There is no circle endpoint among the edges:

$$\neg \exists (i, j) (\{X_i, X_j\} \in \mathcal{E}_{\text{DAG}} \wedge m_{ij} = \circ)$$

- There is no cycle among the graph.

2.2 Fast Causal Inference algorithm (FCI)

Given a database containing n features (equivalently, n variables), the goal of the FCI algorithm is to infer a causal graph that represents the causal relationships underlying the observed data.

Input: An initially complete graph $\mathcal{G}^0$.

Output: A mixed ancestral graph

$$\mathcal{G}_{\text{mixed}} = \max_k \mathcal{G}^k$$

The overall procedure is divided into three phases: *initial skeleton*, *refining skeleton*, and *propagation*, which includes two procedures: *arrow completion* and *tail completion*. The rules of propagation are described in the next section.

During the initial orientation phase, each edge $(X, Y) \in \mathcal{E}$ is associated with a separating set $\text{SepSet}(X, Y)$. At initialization, all separating sets are empty, i.e.,

$$\text{SepSet}(X, Y) \leftarrow \emptyset$$

Algorithm 1 Build the initial skeleton from $\mathcal{G}^0$ [3]

```

 $d \leftarrow 0$ 
repeat
  repeat
    Pick an edge  $\{X_i, X_j\}$  from  $\mathcal{E}^d$ .
    Considering that  $\text{Adj}(X_i)$  denotes the set of adjacent nodes of  $X_i$ .
    for  $\mathbf{Z} \subseteq \text{Adj}(X_i) \setminus \{X_j\}$  where  $|\mathbf{Z}| = d$  do
      if  $X \perp\!\!\!\perp Y \mid \mathbf{Z}$  then
         $\text{SepSet}(X, Y) \leftarrow \text{SepSet}(X, Y) \cup \mathbf{Z}$ 
         $\text{SepSet}(Y, X) \leftarrow \text{SepSet}(Y, X) \cup \mathbf{Z}$ 
         $\mathcal{E}^{d+1} \leftarrow \mathcal{E}^d \setminus \{\{X, Y\}\}$ 
        break
      end if
    end for
  until all edges  $\{X, Y\}$  admitting  $|\text{Adj}(X) \setminus \{Y\}| \geq d$  has been considered.
   $d \leftarrow d + 1$ 
until all edges  $\{X, Y\}$  satisfy  $|\text{Adj}(X) \setminus \{Y\}| < d$ .
return  $\mathcal{G}_1$ 

```

($\mathcal{R}_0$): In $\mathcal{G}_1$, for every unshielded triple $\langle X, Y, Z \rangle$, if $Y \notin \text{SepSet}(X, Z)$, then orient $X * \circ Y \circ * Z$ as $X * \rightarrow Y \leftarrow * Z$. The graph obtained after the sequential application of $\mathcal{R}_0$ is $\mathcal{G}'_1$.

Algorithm 2 Refine the skeleton from $\mathcal{G}'_1$ [3]

```

for  $X_i \in \mathcal{V}$  do
   $pds(X_i) \leftarrow \bigcup_{Y \neq X_i} \text{Possible-D-Sep}(\mathcal{G}'_1, X_i, Y)$ 
  for  $X_j \in \text{Adj}(X_i)$  do
     $d \leftarrow 0$ 
    repeat
      for  $\mathbf{Y} \subseteq pds(X_i) \setminus \{X_j\}$  where  $|\mathbf{Y}| = d$  do
        if  $X_i \perp\!\!\!\perp X_j \mid \mathbf{Y}$  then
           $\text{SepSet}(X_i, X_j) \leftarrow \text{SepSet}(X_i, X_j) \cup \mathbf{Y}$ 
           $\text{SepSet}(X_j, X_i) \leftarrow \text{SepSet}(X_j, X_i) \cup \mathbf{Y}$ 
          Remove  $\{X_i, X_j\}$  from graph.
          break
        end if
      end for
       $d \leftarrow d + 1$ 
    until  $\{X_i, X_j\}$  has already been removed or  $|pds(X_i) \setminus \{X_j\}| < d$ .
    end for
  end for
  Reorient all remaining edges in  $\circ - \circ$ :  $\forall i \forall j \left( \{X_i, X_j\} \in \mathcal{E}^d \implies \begin{cases} m_{ij} \leftarrow \circ \\ m_{ji} \leftarrow \circ \end{cases} \right)$ 
return  $\mathcal{G}_2$ 

```

Possible-D-Sep($\mathcal{G}, X, Y$) returns the set of all nodes V that lie on some path between X and Y such that X and V are connected by at least one inducing path.

While obtaining $\mathcal{G}_2$, the sequential application of $\mathcal{R}_0$ over $\mathcal{G}_2$ occurs to obtain $\mathcal{G}'_2$ at the end.

2.3 Rules of propagation (From $\mathcal{G}'_2$ to $\mathcal{G}_{\text{mixed}}$)

2.3.1 Arrow completion

The sequential application of $\mathcal{R}_0$ replaces several circle endpoints $\circ$ with the arrowhead $>$. However, the replacement is carried out only if it is related to a potential collider, based on the set of separators discovered through the building and refining of the skeleton.

Hence, to guaranty the transitive causality as a significant constraint of a Bayesian network, which is expected to be a Directed Acyclic Graph as the optimal output, several rules are applied over $\mathcal{G}'_2$, which resolve the issues of several remaining circle marks [4].

($\mathcal{R}_1$): For an unshielded triplet $\langle X, Y, Z \rangle$, if $X * \rightarrow Y \circ * Z$ then orient as $Y \rightarrow Z$.

($\mathcal{R}_2$): If $X \rightarrow Y * \rightarrow Z$ or $X * \rightarrow Y \rightarrow Z$ and if $X * \circ Z$ then orient as $X \rightarrow Z$.

($\mathcal{R}_3$): If $X * \rightarrow Y \leftarrow * Z$ and $X * \circ V \circ * Z$ and $V * \circ Y$ and if there is no edge between X and Z then orient as $V * \rightarrow Y$.

($\mathcal{R}_4$): If $\pi = (\pi_k)_{k=1}^l$ is a discriminating path for π_{l-1} and $\pi_{l-1} \circ * \pi_l$ then:

- If $\pi_{l-1} \in \text{SepSet}(\pi_1, \pi_l)$ then orient as $\pi_{l-1} \rightarrow \pi_l$.

- Else, orient as $\pi_{l-2} \leftrightarrow \pi_{l-1} \leftrightarrow \pi_l$.

2.3.2 Tail completion

Besides the efforts to mark the circle extremities with arrowheads, the completeness of the graph still includes the use of tails, which strictly deduces the role of a variable as the main cause of others.

Therefore, by inheriting the four rules of arrow completion mentioned above, Zhang, through an article published in 2008, adds more rules to causal discovery in order to move one step closer to the perfection [5].

($\mathcal{R}_5$): For every remaining unoriented edge $\pi_1 \circ - \circ \pi_l$, if there exists any uncovered circle path $\pi = (\pi_k)_{k=1}^l$ between π_1 and π_l , then orient as $\pi_1 - \pi_l$ and $\forall \pi_k (1 \leq k < l \implies \pi_k - \pi_{k+1})$.

($\mathcal{R}_6$): For any triplet $\langle X, Y, Z \rangle$, if $X - Y \circ - * Z$ then orient as $Y - * Z$.

($\mathcal{R}_7$): For any unshielded triplet $\langle X, Y, Z \rangle$, if $X - \circ Y \circ - * Z$ then orient as $Y - * Z$.

($\mathcal{R}_8$): If $X \rightarrow Y \rightarrow Z$ or $X - \circ Y \rightarrow Z$ and if $X \circ - \rightarrow Z$ then orient as $X \rightarrow Z$.

($\mathcal{R}_9$): If $\pi_1 \circ - \rightarrow \pi_l$ and if there is an uncovered circle path $\pi = (\pi_k)_{k=1}^l$ such that there is no edge between π_2 and π_l then orient as $\pi_1 \rightarrow \pi_l$.

($\mathcal{R}_{10}$): We have a quadruplet of nodes $\langle \alpha, \beta, \gamma, \theta \rangle$ such that:

- $\alpha \circ - \rightarrow \gamma$
- $\beta \rightarrow \gamma \leftarrow \theta$
- There exists an uncovered potentially directed path $p_1 = (\alpha, w_1, \dots, \beta)$
- There exists an uncovered potentially directed path $p_2 = (\alpha, w_2, \dots, \theta)$
- w_1 is possibly β or not.
- w_2 is possibly θ or not.

If w_1 and w_2 are distinct and there is no edge between w_1 and w_2 , then orient as $\alpha \rightarrow \gamma$.

3 Practical implementation

3.1 Choice of libraries

According to the requirements of the subject, the implementation of the FCI algorithm should occur using the `pyagrum` libraries, where we have `pyagrum.BayesNet` for the display and calculation of any Bayesian network, `pyagrum.BNLearner` for calculating tools such as χ^2 or G^2 (used for tests of independence), and `pyagrum.PDAG` for the final representation of the output.

Moreover, we would like to ensure that we implement the algorithm well, the performance of which is proven through the output quality. Hence, the result is initially compared with Carnegie Mellon University's `causal-learn` library, where several causal algorithms, including FCI, are available for further study.

3.2 From $\mathcal{G}_{\text{mixed}}$ to $\mathcal{G}_{\text{PDAG}}$

- **Conventions for imperfect cases**

The output of the FCI algorithm is a Mixed Ancestral Graph (MAG), which may contain several partially oriented edges. Since `pyagrum.PDAG` only supports directed edges and undirected edges, and does not allow edge endpoints with circle marks ($\circ \rightarrow$, $\circ -$) or bidirected edges ($\leftrightarrow$), a set of conventions is applied to map the FCI output to a compatible PDAG representation.

In this version of the algorithm, we provide three different options:

1. We consider $\circ \rightarrow$, $\circ -$ and perfect arrow $\rightarrow$ as one, the others are marked by unoriented edges (in reference to Theorem 2 of Zhang’s article) [5].
2. We consider $\rightarrow$ and $\circ \rightarrow$ as oriented edges for the output, the others become unoriented ones.
3. Only $\rightarrow$ is considered as oriented edge.

- **Risk of cyclic graph**

The application of the above-mentioned conventions may still result in partially directed cycles in the output graph, which causes the code to terminate with an error. To prevent such unexpected termination, we define the following graph construction procedure:

1. Directed edges (arcs) are added first, followed by undirected edges.
2. If adding a directed edge introduces a partially directed cycle, this edge is instead appended to the set of undirected edges.
3. If adding an undirected edge introduces a partially directed cycle, the edge is definitively removed.

The implementation of this procedure relies on a `try--except` control structure to detect and handle cycle-related exceptions during graph construction.

3.3 Meek’s rules (From $\mathcal{G}_{\text{PDAG}}$ to $\mathcal{G}_{\text{DAG}}$)

To obtain a strictly directed acyclic graph from a partially directed acyclic one, according to Zhang’s article, Meek’s algorithm is introduced as a possible solution [5].

($\mathcal{MR}_1$): For any unshielded triplet, if $A \rightarrow B - C$, then orient as $B \rightarrow C$.

($\mathcal{MR}_2$): If $A \rightarrow B \rightarrow C$ and if $A \circ - \circ C$, then orient as $A \rightarrow C$.

($\mathcal{MR}_3$): If $A \rightarrow B \rightarrow C$, $A \circ - \circ D \circ - \circ C$ and if A and C are not adjacent, then orient as $D \rightarrow C$.

The sequential application of those rules is automatically realized with `pyagrum.MeekRules`.

4 Experiments and Results

Source: https://github.com/alexisvuvietnam/MADI_project

This section presents an empirical evaluation of the proposed implementation of the Fast Causal Inference (FCI) algorithm. All experiments are conducted on synthetic datasets generated from random Bayesian networks, with and without artificially introduced latent variables. The performance of FCI is compared to the MIIC algorithm implemented in `pyAgrum`.

4.1 Validation on a Simple Network

We first validate the correctness of the implementation on a small Bayesian network composed of five observed variables. The data are generated from a known Bayesian network structure (Figure 1), allowing a qualitative inspection of the recovered causal graphs.

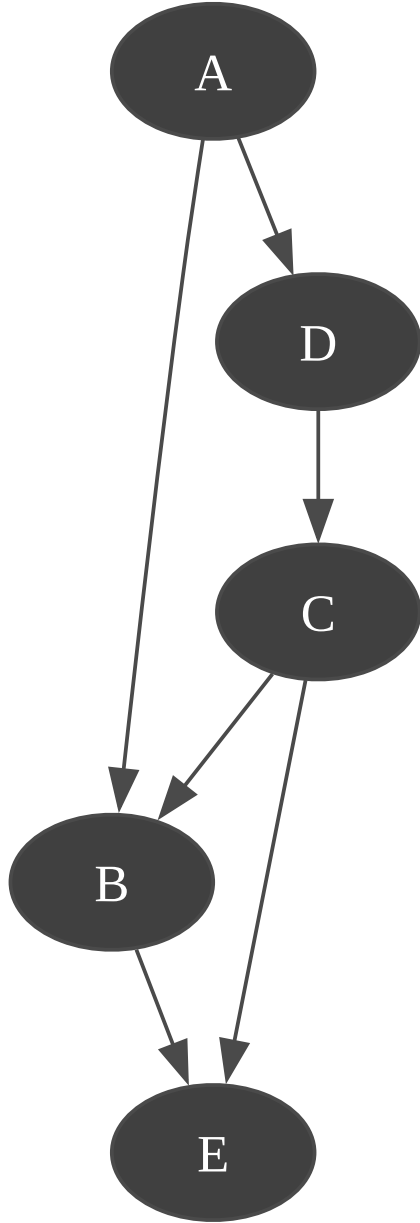


Figure 1: Ground-truth Bayesian network used to generate synthetic data for the simple validation experiment.

Skeleton recovery. FCI is able to recover a plausible skeleton from the data, although no strict ground-truth comparison is possible due to equivalence classes induced by latent variables. The detected skeleton remains stable across runs for this small network.

Edge orientation and uncertainty. Among the five detected edges, all are partially oriented (of the form $X* \rightarrow Y$), but only a small fraction are fully oriented. The resulting certainty score remains low (approximately 17%), highlighting the conservative nature of FCI with respect to causal orientation.

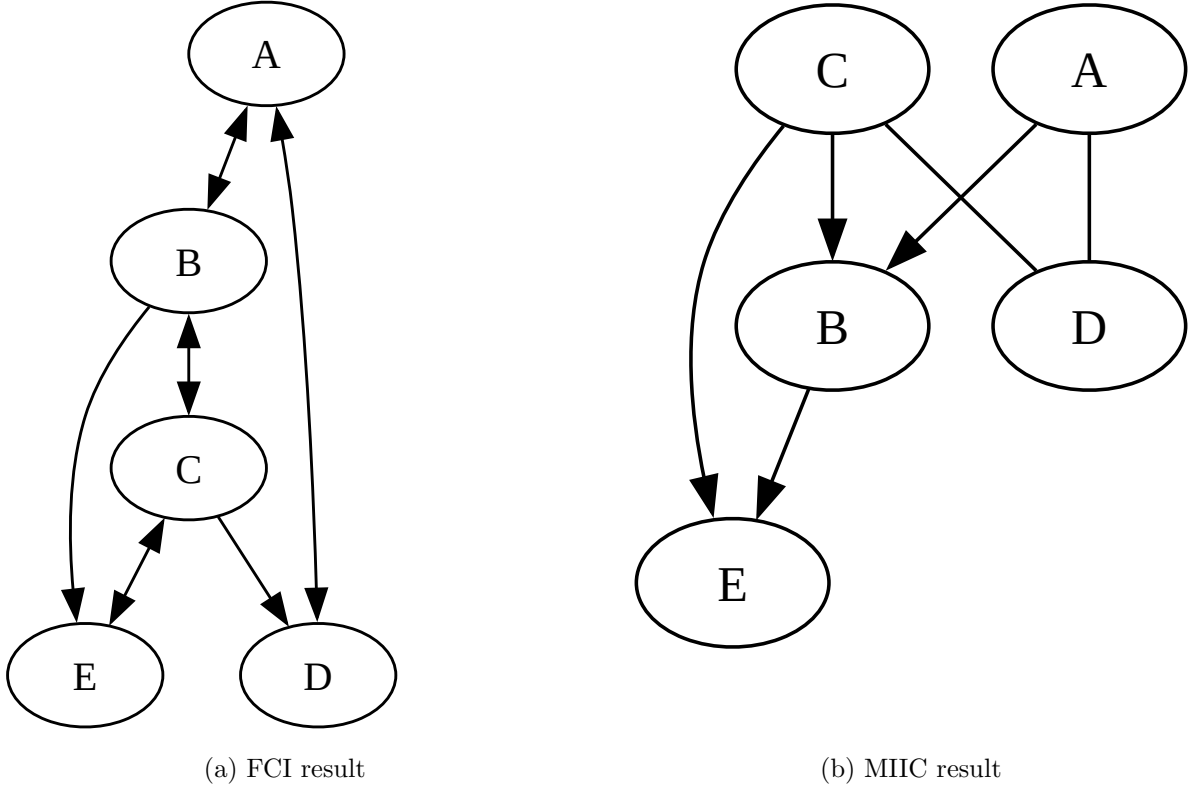


Figure 2: Comparison between FCI and MIIC on the same dataset.

Latent variable simulation. Latent confounding is simulated by generating data from the full Bayesian network and subsequently removing one variable at random from the dataset. While this approach introduces hidden common causes, it may over-simplify real-world latent structures and can lead to instability in the estimated separating sets.

Separating sets. The separating sets identified by FCI show significant variability across different generated datasets, even when the underlying Bayesian network remains fixed. This sensitivity is mainly due to sampling variability and the limited sample size.

Overall, execution time remains negligible for networks of four to five variables.

4.2 Comparison Between FCI and MIIC

To enable a fair comparison, MIIC is used to learn a partially directed acyclic graph (PDAG), which is closer in spirit to the Partial Ancestral Graph (PAG) produced by FCI.

Skeleton similarity. Both algorithms tend to recover similar skeletons, although discrepancies may appear depending on the sampled dataset. The Jaccard similarity coefficient ranges between 0.6 and 0.9 for most configurations (Figure 4).

Edge orientation. MIIC generally produces a larger number of oriented edges than FCI. This behavior is consistent both in the standard setting and in the presence of latent variables, where FCI often leaves edges unoriented to reflect causal uncertainty.

Computational performance. MIIC consistently outperforms FCI in terms of execution time, being between 3 and 20 times faster depending on the network size and sample size (Figure ??).

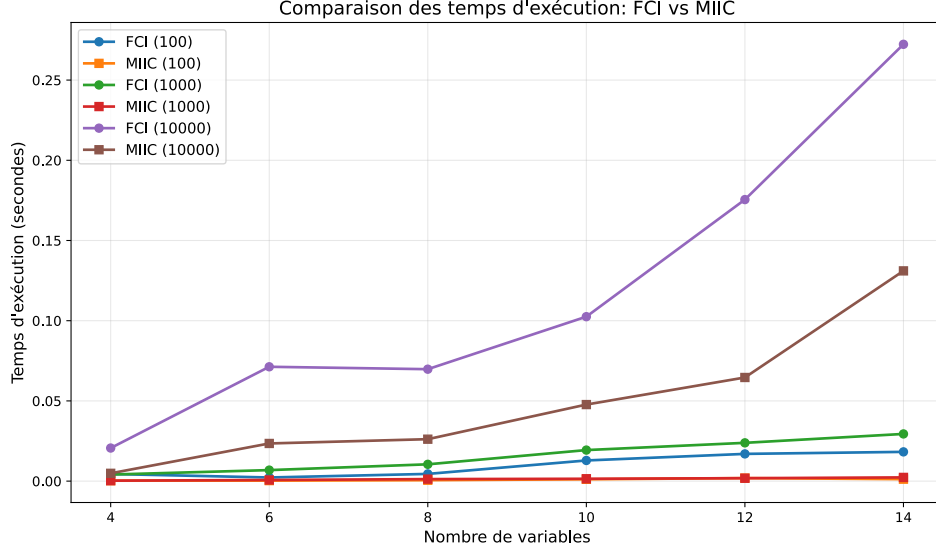


Figure 3: Execution time of FCI and MIIC as a function of network size and sample size.

Summary. Both algorithms converge on the global structure of the network. However, FCI explicitly encodes causal uncertainty through partially oriented edges, while MIIC favors more aggressive orientation at the cost of potential false positives.

4.3 Impact of Sample Size

The influence of the sample size is evaluated for $n \in \{100, 1000, 10000\}$ observations.

Small sample size ($n = 100$). The recovered skeleton is unstable and often either incomplete or overly dense. FCI exhibits a high uncertainty score, with many circle endpoints. The Jaccard similarity between FCI and MIIC is generally low (Figure 4).

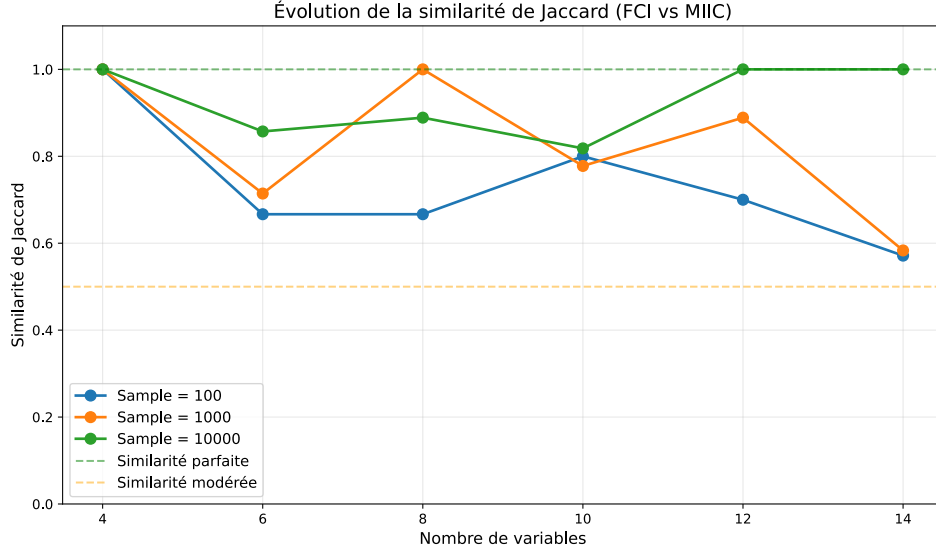


Figure 4: Jaccard similarity between the skeletons inferred by FCI and MIIC for different network sizes and sample sizes.

Medium sample size ($n = 1000$). Skeleton recovery becomes more stable and edge orientations are more reliable. The variability in the number of detected edges decreases significantly, and the uncertainty score is reduced (Figure 5).

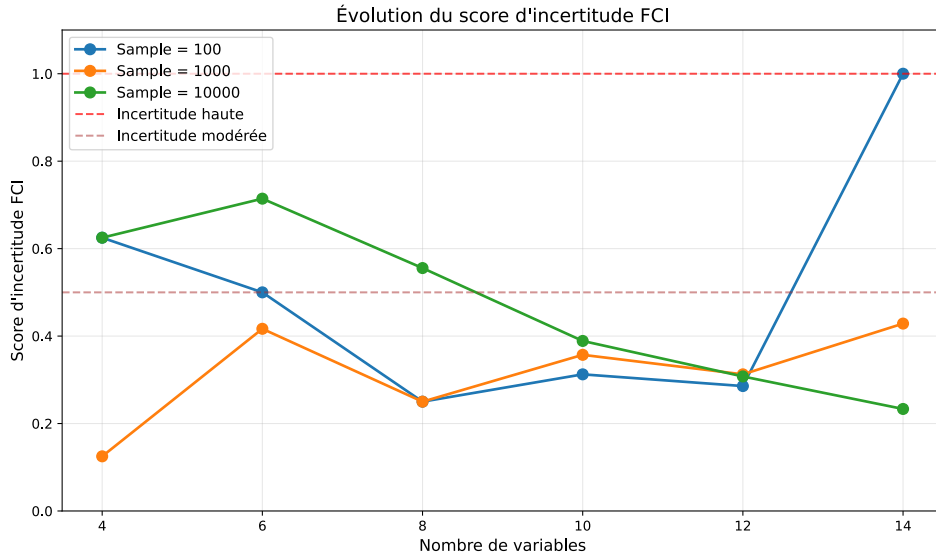


Figure 5: Evolution of the FCI uncertainty score as a function of network size and sample size.

Large sample size ($n = 10000$). Most orientations become stable and the uncertainty score drops below 0.4 for the majority of networks. The Jaccard similarity exceeds 0.8 in most cases, indicating strong agreement between the two algorithms.

Conclusion. Sample size is a critical factor for causal discovery. FCI requires substantially more data than MIIC to produce reliable orientations, and improvements beyond 1000 samples

tend to become marginal.

4.4 Impact of Network Size

All experiments in this section are conducted with $n = 10000$ samples unless stated otherwise.

Execution time. Execution time increases rapidly with the number of variables, especially for FCI, reflecting its higher number of conditional independence tests. The ratio between FCI and MIIC execution times decreases for larger networks, although FCI remains consistently slower (Figure 6).

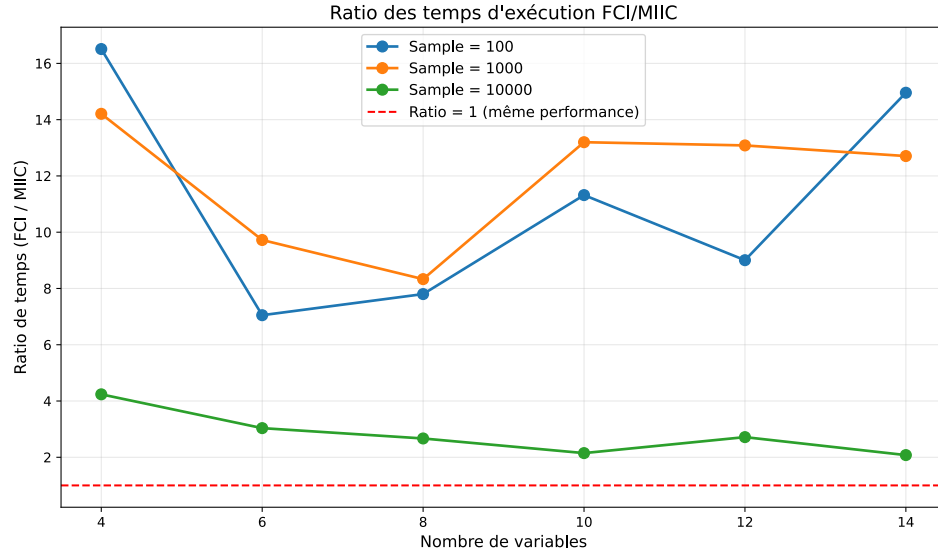


Figure 6: Ratio between FCI and MIIC execution times as a function of network size for different sample sizes.

Structural differences. MIIC generally detects more edges than FCI, particularly for larger networks. The difference in the number of detected edges (MIIC minus FCI) increases with network size, indicating a more aggressive behavior of MIIC (Figure 7).

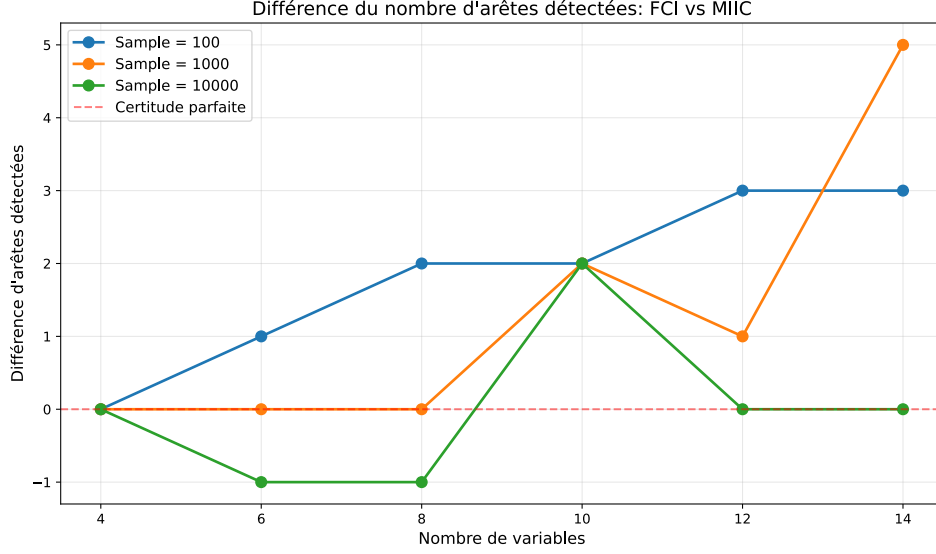


Figure 7: Difference in the number of detected edges between MIIC and FCI (MIIC minus FCI) as a function of network size.

Jaccard similarity. The similarity between the two algorithms decreases as the number of variables increases, especially for small sample sizes. Nevertheless, similarity values remain acceptable (above 0.5) in most configurations.

FCI uncertainty. The uncertainty score produced by FCI decreases with increasing network size but remains strongly dependent on sample size. Scores above 0.5 indicate that the inferred causal relations should be interpreted with caution (Figure 5).

4.5 Sensitivity to the Significance Level α

The sensitivity of FCI to the conditional independence significance level α is evaluated on a network with ten variables and 10000 samples.

Strict threshold ($\alpha = 0.001$). Very few edges are detected, leading to sparse skeletons and a high risk of false negatives.

Standard threshold ($\alpha = 0.05$). Results remain similar to the strict setting, while execution time is slightly reduced. This value provides a reasonable trade-off between precision and recall.

Permissive threshold ($\alpha = 0.2$). The inferred skeleton becomes dense, uncertainty increases, and the risk of false positives is higher. Execution time also increases due to a larger number of candidate tests (Figure 8).

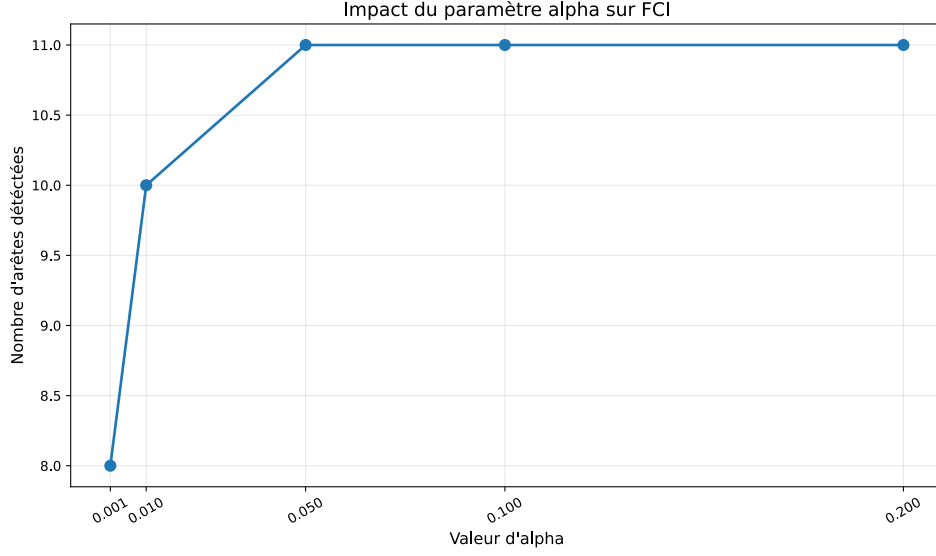


Figure 8: Impact of the significance level α on the number of detected edges.

Conclusion. The choice of α strongly affects the balance between precision and recall. While $\alpha = 0.05$ appears to be a good default choice, the optimal value depends on data availability and the desired level of conservativeness.

4.6 Overall Conclusions

The experiments confirm that the proposed implementation of FCI is functionally correct and behaves consistently with theoretical expectations. Compared to MIIC, FCI is computationally more expensive but provides a more faithful representation of causal uncertainty, particularly in the presence of latent variables. While scalability remains limited for large networks, FCI is well-suited for robust causal discovery when sufficient data are available.

5 Annexe

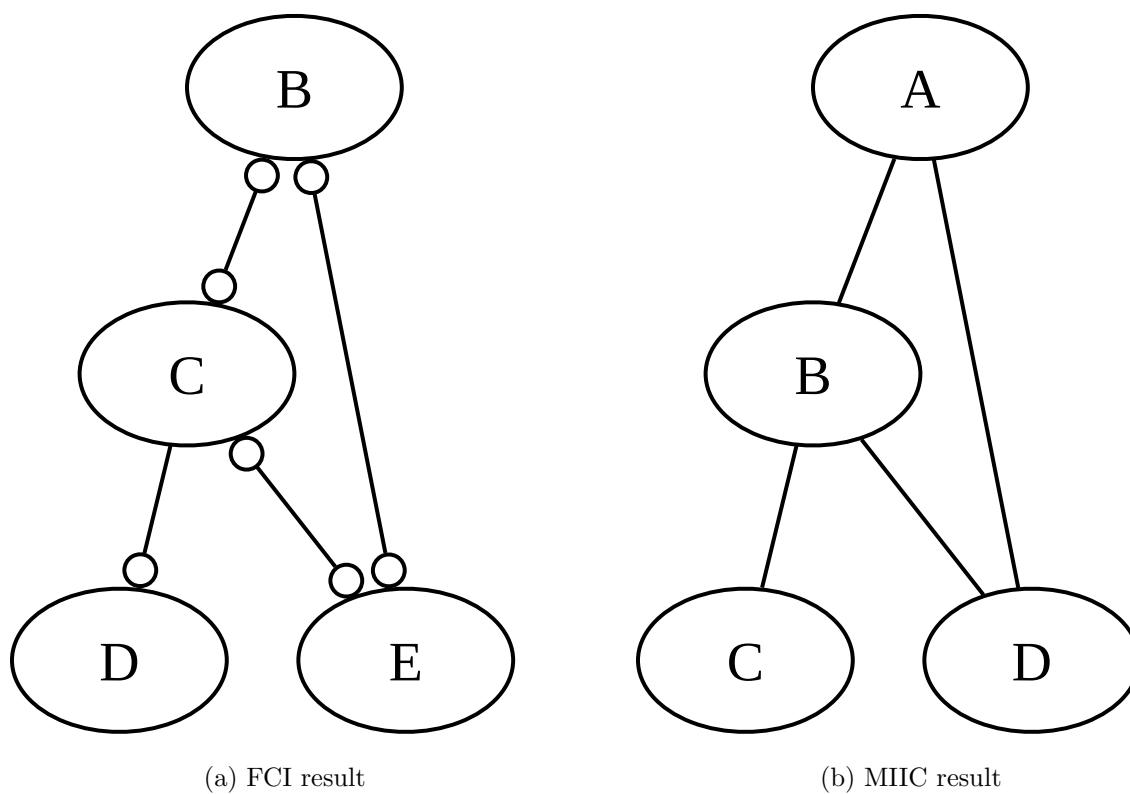


Figure 9: Comparison between FCI and MIIC on the same dataset with a simulated latent variable.

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